FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

BSc Data Science & AI - Semester 2

COMPREHENSIVE REVISION NOTE 4

Probability, Bayes' Rule & Game Theory

TOPICS COVERED IN THIS REVISION NOTE

1. Introduction to Uncertainty in AI
2. Probability Theory Fundamentals
3. Conditional Probability
4. Bayes' Theorem - Derivation and Meaning
5. Prior, Likelihood, and Posterior Probabilities
6. Applications of Bayes' Rule
7. Introduction to Game Theory in AI
8. Adversarial Search
9. Minimax Algorithm
10. Alpha-Beta Pruning
11. Evaluation Functions

SECTION 1: DEALING WITH UNCERTAINTY IN AI

1.1 Why Uncertainty Matters

In the real world, agents rarely have complete and certain knowledge about their environment. Unlike the deterministic, rule-based systems we studied with propositional logic, real-world AI systems must deal with:

- **Incomplete information:** The agent doesn't have access to all relevant facts
- **Noisy sensors:** Measurements and observations contain errors
- **Unpredictable outcomes:** Actions may have uncertain effects
- **Dynamic environments:** The world changes in ways the agent cannot predict

Probability theory provides a mathematically rigorous framework for reasoning under uncertainty. It allows us to quantify our beliefs about the world and update them as new evidence arrives. This is fundamentally different from logic, which deals with absolute truth and falsity.

LOGIC vs PROBABILITY

Logic (Deterministic):
- Statements are TRUE or FALSE
- Rules are certain: "If P then Q"
- Cannot handle uncertainty
- Good for well-defined domains

Probability (Stochastic):
- Statements have degrees of belief (0 to 1)
- Rules express likelihood: "P makes Q more probable"
- Explicitly handles uncertainty
- Essential for real-world AI applications

SECTION 2: PROBABILITY THEORY FUNDAMENTALS

2.1 Basic Definitions

Probability measures the likelihood of an event occurring. It is a number between 0 (impossible) and 1 (certain). We can think of probability as the "degree of belief" that something is true.

Term	Symbol	Definition	Example
Sample Space	Ω	Set of all possible outcomes	Coin flip: Ω = {H, T}
Event	A, B, E	A subset of the sample space	Getting heads: A = {H}
Probability	P(A)	Number between 0 and 1	P(Heads) = 0.5
Complement	A' or ¬A	Event A does NOT occur	P(¬Heads) = 0.5
Union	A ∪ B	A OR B (at least one)	Roll 1 OR 2 on die
Intersection	A ∩ B	A AND B (both occur)	Even AND greater than 3

2.2 Axioms of Probability

■ **THE THREE AXIOMS OF PROBABILITY**

Axiom 1 (Non-negativity): For any event A, $P(A) \geq 0$
Probabilities cannot be negative.

Axiom 2 (Normalization): $P(\Omega) = 1$
The probability of the entire sample space is 1 (something must happen).

Axiom 3 (Additivity): For mutually exclusive events A and B:
$P(A \cup B) = P(A) + P(B)$
If events cannot both happen, their combined probability is the sum.

Key Consequence: $P(A) + P(\neg A) = 1$ (complementary events)

2.3 Important Probability Rules

Rule Name	Formula	When to Use
Complement Rule	$P(\neg A) = 1 - P(A)$	Find P(not A) from P(A)
Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Find P(A or B) for any events
Multiplication Rule	$P(A \cap B) = P(A\|B) \times P(B)$	Find P(A and B) using conditional
Independence	$P(A \cap B) = P(A) \times P(B)$	When A and B don't affect each other

SECTION 3: CONDITIONAL PROBABILITY

3.1 What is Conditional Probability?

Conditional probability is the probability of an event A occurring GIVEN that another event B has already occurred. It answers the question: "How does knowing B change my belief about A?".

$$P(A|B) = P(A \cap B) / P(B) \text{ where } P(B) > 0$$

This formula captures the intuition that when we know B has occurred, our sample space shrinks from Ω to just B. The conditional probability is then the fraction of B that also belongs to A.

■ UNDERSTANDING CONDITIONAL PROBABILITY VISUALLY

Think of it as restricting your focus:

Without condition: P(A) asks "What fraction of the total sample space is A?"

With condition: P(A|B) asks "Given we're already in B, what fraction of B is also in A?"

Example: Die roll
- P(6) = 1/6 (one of six faces)
- P(6 | Even) = P(6 ∩ Even) / P(Even) = (1/6) / (3/6) = 1/3
- Knowing the roll is even reduces sample space to {2, 4, 6}, so P(6) becomes 1/3

3.2 The Multiplication Rule

From the definition of conditional probability, we can derive the **multiplication rule**:

$$P(A \cap B) = P(A|B) \times P(B) = P(B|A) \times P(A)$$

This tells us that the probability of both A and B occurring equals the probability of B times the probability of A given B. This rule is fundamental to Bayes' theorem.

3.3 Law of Total Probability

If events B■, B■, ..., B■ partition the sample space (they're mutually exclusive and exhaustive), then:

$$P(A) = P(A|B\blacksquare)P(B\blacksquare) + P(A|B\blacksquare)P(B\blacksquare) + \ldots + P(A|B\blacksquare)P(B\blacksquare)$$

Special case with two events: If B and ¬B partition the space:

$$P(A) = P(A|B)P(B) + P(A|\neg B)P(\neg B)$$

SECTION 4: BAYES' THEOREM

4.1 Deriving Bayes' Theorem

Bayes' Theorem is one of the most important results in probability and the foundation of Bayesian reasoning in AI. It tells us how to update our beliefs when we receive new evidence.

■ **DERIVATION OF BAYES' THEOREM**

Start with conditional probability:
$P(A|B) = P(A \cap B) / P(B)$ ... (1)

Also, from the multiplication rule:
$P(A \cap B) = P(B|A) \times P(A)$ ... (2)

Substitute (2) into (1):

■ **BAYES' THEOREM**

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Or equivalently, using the law of total probability:

$$P(A|B) = P(B|A) \times P(A) / [P(B|A) \times P(A) + P(B|\neg A) \times P(\neg A)]$$

4.2 Understanding the Components

Component	Symbol	Name	Meaning
P(A|B)	Posterior	Posterior Probability	Updated belief in A after seeing evidence B
P(B|A)	Likelihood	Likelihood	How likely we'd see B if A were true
P(A)	Prior	Prior Probability	Initial belief in A before evidence
P(B)	Evidence	Total Probability of Evidence	How likely the evidence is overall

4.3 The Bayesian Interpretation

Bayes' theorem provides a principled way to **update beliefs based on evidence**. Think of it as:

Posterior ∝ Likelihood × Prior

Your updated belief (posterior) is proportional to how well the evidence fits your hypothesis (likelihood) times your initial belief (prior). The evidence P(B) acts as a normalizing factor to ensure probabilities sum to 1.

■ **BAYESIAN INTUITION**

How Bayes' Rule Models Rational Belief Update:

1. **Start with a Prior:** Before seeing any evidence, you have some initial belief P(A). This could come from past experience, domain knowledge, or reasonable assumptions.

2. **Observe Evidence:** You see evidence B (a test result, a symptom, a data point).

3. **Ask "How likely is this evidence?":** If A were true, how likely would we see B? This is P(B|A), the likelihood.

4. **Update your belief:** Your new belief P(A|B) combines your prior with how well the evidence fits. Strong evidence that fits hypothesis A well will increase your belief in A. Evidence that doesn't fit A will decrease your belief.

SECTION 5: APPLYING BAYES' THEOREM

5.1 Medical Diagnosis Example

■ **WORKED EXAMPLE: MEDICAL TESTING**

Problem: A disease affects 1% of the population. A test for the disease is 95% accurate:
• If you have the disease, test is positive 95% of the time (sensitivity)
• If you don't have the disease, test is negative 95% of the time (specificity)

Question: If you test positive, what's the probability you actually have the disease?

Given Information:
• P(Disease) = 0.01 (prior - 1% have disease)
• P(No Disease) = 0.99
• P(Positive | Disease) = 0.95 (likelihood - true positive rate)
• P(Positive | No Disease) = 0.05 (false positive rate)

Apply Bayes' Theorem:
P(Disease | Positive) = P(Positive | Disease) × P(Disease) / P(Positive)

Calculate P(Positive) using Law of Total Probability:
P(Positive) = P(Positive|Disease)×P(Disease) + P(Positive|No Disease)×P(No Disease)
P(Positive) = (0.95 × 0.01) + (0.05 × 0.99) = 0.0095 + 0.0495 = 0.059

Final Calculation:
P(Disease | Positive) = (0.95 × 0.01) / 0.059 = 0.0095 / 0.059 ≈ **0.161 or 16.1%**

Surprising Result! Even with a positive test, there's only a 16% chance you have the disease! This is because the disease is rare (1%), so most positive tests are false positives.

5.2 Key Insights from Bayes' Theorem

• **Base Rate Matters:** The prior probability significantly affects the posterior. Rare events remain somewhat unlikely even with positive evidence.

• **Evidence Updates Beliefs:** Strong evidence (high likelihood ratio) can overcome a low prior.

• **Multiple Evidence:** Bayes' rule can be applied iteratively - each piece of evidence updates the prior.

SECTION 6: GAME THEORY IN AI

6.1 Introduction to Game Theory

Game theory is the study of strategic decision-making in situations where the outcome depends not just on your actions but also on the actions of other agents (players). In AI, game theory is essential for:

- Building game-playing programs (chess, Go, poker)
- Multi-agent systems where agents have conflicting goals
- Adversarial scenarios (security, negotiation)

6.2 Types of Games

Property	Type A	Type B	Examples
Information	Perfect: All info visible	Imperfect: Hidden info	Chess (perfect) vs Poker (imperfect)
Sum	Zero-sum: One wins, other loses	Non-zero: Both can gain/lose	Chess (zero) vs Trade (non-zero)
Turns	Sequential: Take turns	Simultaneous: Act at once	Chess (seq) vs Rock-Paper-Scissors (sim)
Outcome	Deterministic: No chance	Stochastic: Involves randomness	Chess (det) vs Backgammon (stoch)

6.3 Adversarial Search

In games like chess or tic-tac-toe, we need **adversarial search** algorithms that assume the opponent plays optimally against us. Unlike regular search where we control all actions, here we must plan for an opponent who actively tries to minimize our success.

The game is represented as a **game tree** where nodes are game states and edges are moves. Players alternate turns, and leaf nodes have utility values (who wins/loses).

SECTION 7: THE MINIMAX ALGORITHM

7.1 Core Idea

The **Minimax algorithm** is the foundation of game-playing AI. It operates on a simple but powerful principle: assume your opponent plays perfectly, and choose moves that maximize your minimum guaranteed outcome.

> ■ **MINIMAX PRINCIPLE**
>
> **MAX player (us):** Wants to MAXIMIZE the utility (score)
> **MIN player (opponent):** Wants to MINIMIZE our utility (opponent plays against us)
>
> **Strategy:**
> • At MAX nodes (our turn): Choose the move with the HIGHEST minimax value
> • At MIN nodes (opponent's turn): Assume they choose the move with the LOWEST minimax value
> • At terminal nodes: Use the actual utility value (who won/lost)
>
> **Key Insight:** We don't hope the opponent makes mistakes. We plan for the worst case - a perfect opponent - and still try to do our best.

7.2 Minimax Algorithm Steps

MINIMAX ALGORITHM

```
function MINIMAX(state, depth, isMaximizing):

  if depth = 0 or state is terminal:
    return evaluate(state) // utility value

  if isMaximizing: // MAX player's turn
    maxEval = -∞
    for each child of state:
      eval = MINIMAX(child, depth-1, FALSE)
      maxEval = max(maxEval, eval)
    return maxEval

  else: // MIN player's turn
    minEval = +∞
    for each child of state:
      eval = MINIMAX(child, depth-1, TRUE)
      minEval = min(minEval, eval)
    return minEval
```

7.3 Minimax Properties

Property	Value	Explanation
Complete	Yes (for finite trees)	Will find a solution if game tree is finite
Optimal	Yes (against optimal opponent)	Finds best move assuming perfect play from opponent

Time Complexity	O(b^m)	b = branching factor, m = max depth
Space Complexity	O(bm)	Depth-first exploration

SECTION 8: ALPHA-BETA PRUNING

8.1 The Problem with Minimax

Pure minimax is exponentially expensive - it explores the entire game tree. For chess with ~35 possible moves per turn and games lasting ~80 moves, we'd need to explore 35^80 positions - impossible! **Alpha-Beta pruning** is an optimization that eliminates branches that cannot possibly affect the final decision.

8.2 How Alpha-Beta Works

■ **ALPHA-BETA PRUNING CONCEPT**

Alpha (α): The best (highest) value that MAX can guarantee so far
Beta (β): The best (lowest) value that MIN can guarantee so far

Pruning Rule:
• At a MIN node: If value ≤ α, prune remaining children (MAX already has a better option)
• At a MAX node: If value ≥ β, prune remaining children (MIN already has a better option)

Intuition:
If you've found a move that gives you at least value α, you don't need to explore moves that your opponent can force to be worse than α. Similarly, your opponent won't let you have more than β.

8.3 Alpha-Beta Benefits

- **Same Result:** Alpha-beta never changes the minimax decision - it just finds it faster
- **Pruning Efficiency:** With perfect ordering, time complexity reduces from O(b^m) to O(b^(m/2))
- **Move Ordering:** Examining better moves first leads to more pruning
- **Effective Depth:** Can search roughly twice as deep in the same time

8.4 Evaluation Functions

Since we often cannot search to terminal states, we use **evaluation functions** to estimate the utility of non-terminal states. A good evaluation function should:

- Return the actual utility at terminal states
- Be quick to compute (we call it millions of times)
- Correlate strongly with actual chance of winning
- Consider material balance, position, mobility, king safety (for chess)

EXAMPLE: SIMPLE CHESS EVALUATION

Material Values: Pawn=1, Knight=3, Bishop=3, Rook=5, Queen=9

Basic Evaluation:
E(s) = Σ(piece_value × piece_count_white) - Σ(piece_value × piece_count_black)

Real chess programs add: pawn structure, king safety, piece activity, center control, etc.

SECTION 9: SAMPLE QUIZ QUESTIONS

9.1 Multiple Choice Questions

1. In Bayes' theorem, P(A) is called the:

a) Posterior
b) Likelihood
c) Prior
d) Evidence
Answer: c) Prior (initial belief before evidence)

2. In minimax, what does the MAX player try to do?

a) Minimize utility
b) Maximize utility
c) Minimize opponent's moves
d) Draw the game
Answer: b) Maximize utility

3. Alpha-beta pruning:

a) Changes the minimax result
b) Always explores all nodes
c) Reduces time while giving same result
d) Only works for chess
Answer: c) Reduces time while giving same result

4. If $P(A) = 0.3$, what is $P(\neg A)$?

a) 0.3
b) 0.7
c) 0.5
d) Cannot determine
Answer: b) 0.7 (complement rule: 1 - 0.3)

5. In alpha-beta, α represents:

a) MIN's best guarantee
b) MAX's best guarantee so far
c) The evaluation function
d) The depth limit
Answer: b) MAX's best (highest) value guaranteed so far

6. $P(A|B) \times P(B)$ equals:

a) $P(A)$
b) $P(B|A)$
c) $P(A \cap B)$
d) $P(A \cup B)$
Answer: c) $P(A \cap B)$ (multiplication rule)

7. A zero-sum game means:

a) Both players lose
b) One player's gain equals other's loss
c) No one wins
d) Game always ends in draw
Answer: b) One player's gain equals other's loss

8. In the medical testing example, why is P(Disease|Positive) often lower than expected?

a) Tests are inaccurate

b) Doctors make mistakes
c) Low prior probability (rare disease)
d) Math is wrong
Answer: c) Low prior probability - base rate matters!

9.2 True/False Questions

1. $P(A \cap B) = P(A) \times P(B)$ always holds. **(FALSE (only for independent events))**

2. Bayes' theorem allows us to find $P(A|B)$ from $P(B|A)$. **(TRUE)**

3. In minimax, we assume the opponent plays randomly. **(FALSE (assume optimal play))**

4. Alpha-beta pruning can search deeper than pure minimax in same time. **(TRUE (effectively doubles depth))**

5. $P(A|B) = P(B|A)$ always. **(FALSE (different quantities))**

6. The sum of all probabilities in a sample space equals 1. **(TRUE (normalization axiom))**

7. Chess is a perfect information game. **(TRUE (all information visible))**

8. Evaluation functions are only used at terminal nodes. **(FALSE (used at cutoff depth for non-terminal nodes))**

SECTION 10: KEY FORMULAS SUMMARY

ESSENTIAL FORMULAS TO REMEMBER

PROBABILITY BASICS:
- $P(\neg A) = 1 - P(A)$ (Complement)
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (Addition)
- $P(A \cap B) = P(A|B) \times P(B)$ (Multiplication)
- If independent: $P(A \cap B) = P(A) \times P(B)$

CONDITIONAL PROBABILITY:
- $P(A|B) = P(A \cap B) / P(B)$ where $P(B) > 0$

BAYES' THEOREM:
- $P(A|B) = P(B|A) \times P(A) / P(B)$
- $P(A|B) = P(B|A) \times P(A) / [P(B|A)\times P(A) + P(B|\neg A)\times P(\neg A)]$
- Posterior = (Likelihood × Prior) / Evidence

LAW OF TOTAL PROBABILITY:
- $P(A) = P(A|B)\times P(B) + P(A|\neg B)\times P(\neg B)$

MINIMAX:
- MAX nodes: value = max(children values)
- MIN nodes: value = min(children values)
- Time: $O(b^m)$, Space: $O(bm)$

ALPHA-BETA:
- α = MAX's best guarantee (prune when value $\leq \alpha$ at MIN)
- β = MIN's best guarantee (prune when value $\geq \beta$ at MAX)
- Best case time: $O(b^{(m/2)})$

— End of Revision Note 4 —
Best of luck with your exam preparation!

This completes the Foundations of AI revision notes series.